

ETHICS

What are ethics?

- *Ethics are a structure of standards, values, and practices that influence how people lead their lives.*
- *Not law*

Computer ethics

are a structure of standards, values, and practices that influence how people should process and consume computing technology and its related disciplines without damaging or violating the moral values and beliefs of any individual, organisation or entity.

Why are ethics important?

They ensure we:

- contribute to the wellbeing of the society acknowledging that all people are stakeholders in computing.
- build fair technology that does not to discriminate
- respect users' privacy and honour confidentiality.
- engage users in a respectable and safe manner
- respect other people's ideas and artefacts

Why do we need ethics for security practitioners?

- Security researchers are put in ***position of trust***
- Security jobs and roles are usually ***out of the public eye*** (scrutiny)
- Security researchers poses ***special knowledge and skills*** which are not common to everyone
- Security researchers may be in position of **power** with their clients, **power dynamics**.
- Security work or research could lead to public ***harm*** in some cases ***loss of life***.

What makes a good code?

- Detailed guidance on how to interpret and apply principles
- Addresses the relationship between entities
 - practitioner and client
 - Practitioner and society
 - (Balancing the relationships)
- Adoption and support by a well-defined community of practitioners

Examples

- ACM Code of ethics and Professional Conduct (2018)
- CREST code of Conduct

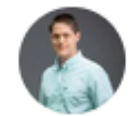
ACM Code of Ethics and Professional Conduct

GENERAL ETHICAL PRINCIPLES

A computing professional should...

- 1.1 Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing.
- 1.2 Avoid harm.
- 1.3 Be honest and trustworthy.
- 1.4 Be fair and take action not to discriminate.
- 1.5 Respect the work required to produce new ideas, inventions, creative works, and computing artifacts.
- 1.6 Respect privacy.
- 1.7 Honour confidentiality.

Case study



James Davis

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Ethical conduct in cybersecurity research

What happened?

Researchers at the University of Minnesota conducted a research project titled *On the Feasibility of Stealthily Introducing Vulnerabilities in Open-Source Software via Hypocrite Commits*.

2018

UMN bug-fix research on Linux kernel starts, and roughly 400 bug-fix patches are contributed over the next two years, mainly centered around specific research papers

August 2020

"Hypocrite Commits" patches from UMN researchers sent to kernel developers under false identities.

November 2020

(1) "Hypocrite Commits" paper is published + accepted by IEEE Symposium on Security and Privacy

(2) Sarah Jamie Lewis calls attention to paper's ethics

- Is the threat model of hypocrite commits correct?
- Was this really a "safe" proof-of-concept?
- (When) are these methods for academic research sound?
- Should these methods be applied on cybersecurity?
- Should there be "no-go" ethical boundaries for safety-critical projects, such as the linux kernel?

Further Reading

<https://www.nytimes.com/2018/04/04/us/politics/cambridge-analytica-scandal-fallout.html>

<https://www.theguardian.com/technology/2019/mar/17/the-cambridge-analytica-scandal-changed-the-world-but-it-didnt-change-facebook>

Predicting whether people will lie online: <https://dl.acm.org/doi/pdf/10.1145/3411764.3445625>

Hypocrite commits paper disclosure: <https://www-users.cse.umn.edu/~kjl/papers/full-disclosure.pdf>

Hypocrite commits full paper: <https://linuxreviews.org/images/d/d9/OpenSourceInsecurity.pdf>